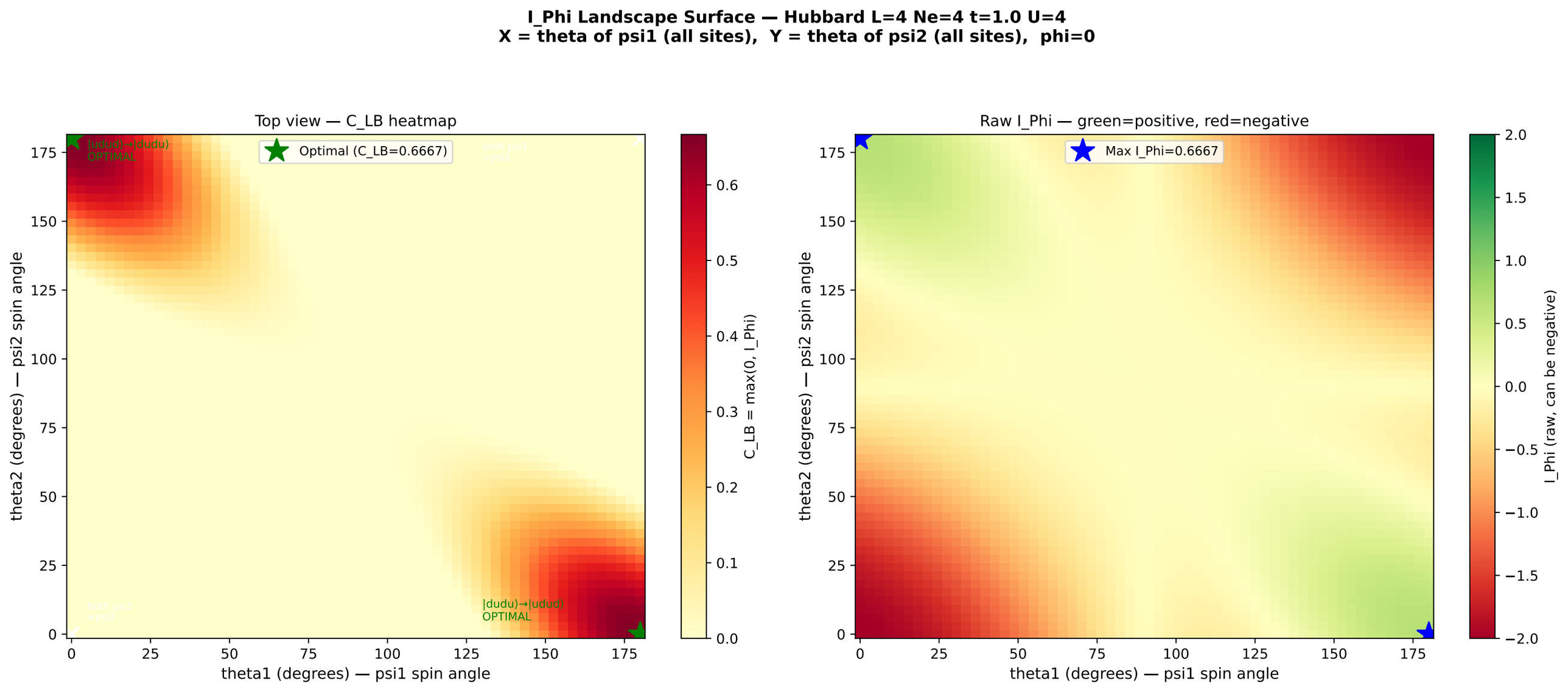
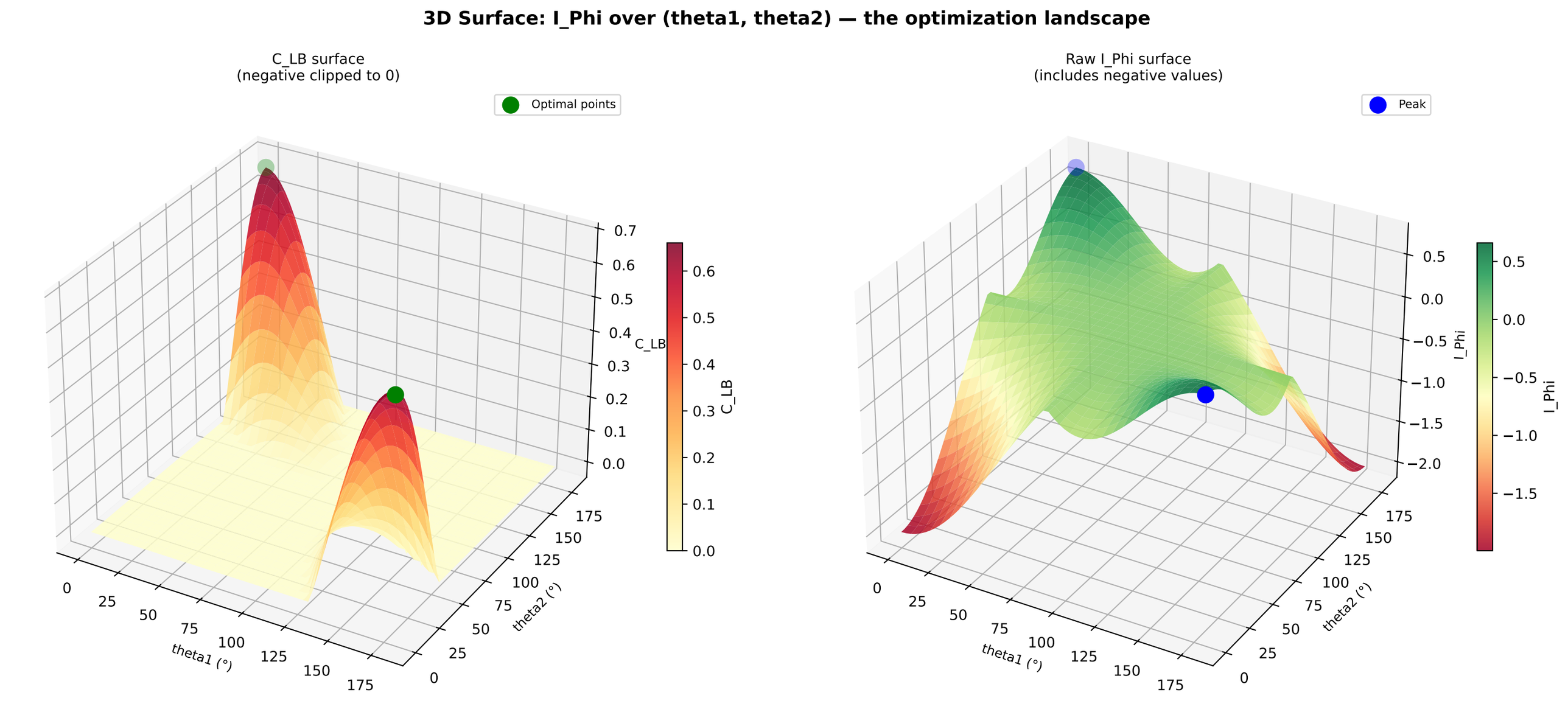

I_Phi Landscape Surface — Hubbard L=4 Ne=4 t=1.0 U=4
X = theta of psi1 (all sites), Y = theta of psi2 (all sites), phi=0
Top view — C_LB heatmap
Optimal (C_LB=0.6667)
|ududu)→|dudu)
OPTIMAL
both psi1
=psi2
|dudu)→|ududu)
OPTIMAL
both psi1
=psi2
theta2 (degrees) — psi2 spin angle
theta1 (degrees) — psi1 spin angle
C_LB = max(0, I_Phi)
Raw I_Phi — green=positive, red=negative
Max I_Phi=0.6667
theta2 (degrees) — psi2 spin angle
theta1 (degrees) — psi1 spin angle
I_Phi (raw, can be negative)


3D Surface: I_Phi over (theta1, theta2) — the optimization landscape
C_LB surface
(negative clipped to 0)
Optimal points
C_LB
theta1 (°)
theta2 (°)
Raw I_Phi surface
(includes negative values)
Peak
I_Phi
theta1 (°)
theta2 (°)